



FINOPTIX

SUMMER PROJECT '25



Assignment - 4 | Black Litterman Model

Assignment Structure

Objective:

In this assignment, you will implement the Black-Litterman model using Python for a portfolio of 20 large-cap US stocks. The goal is to understand how market equilibrium and investor views can be combined to generate improved estimates for portfolio optimization.

In this assignment, you have been provided with a Python template to implement the Black-Litterman model with some missing code cells. You are required to complete the missing code so that the implementation runs correctly and submit the completed Python notebook.

Link to Python Notebook → https://colab.research.google.com/drive/1Mk4NIEEbcIbhl_diqz1hx7aeMvNXSho0?usp=sharing

Dataset and Setup:

- Use daily adjusted close prices from 2020-01-01 to 2024-12-31.
- Use the yfinance library to fetch the data.
- Stocks to be included (20 total): AAPL, MSFT, GOOGL, AMZN, META, TSLA, NVDA, JPM, JNJ, PG, V, MA, HD, DIS, PFE, KO, PEP, ADBE, INTC, NFLX.

Tasks:

Part A: Preprocessing

1. Fill in the missing code to compute log returns.
2. Drop any stocks with incomplete data.

Part B: Market Equilibrium (Prior)

1. Implement the implied_returns(...) function using: $\pi = \delta \times \Sigma \times w_mkt$
 - δ : risk aversion coefficient (use 2.5)
 - Σ : sample covariance matrix
 - w_mkt : equal weight portfolio (as simplification)

Part C: Investor Views

1. Fill the P matrix for the 3 views given:
 - AAPL will outperform MSFT by 2%
 - TSLA will underperform NVDA by 3%
 - AMZN will outperform DIS by 1.5%
2. Define the Q vector with the expected return differences.
3. Fill the Omega matrix: diagonal with small values (e.g., 0.0001) indicating view uncertainty.

Part D: Posterior Calculation

1. Complete the function black_litterman_posterior(...):
 - Use the formula:
 - $\mu_bl = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1} \times [(\tau\Sigma)^{-1}\pi + P^T\Omega^{-1}Q]$
 - $\Sigma_bl = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}$

Part E: Portfolio Optimization

1. Implement mean_variance_opt(...):
 - Use optimization (e.g. scipy.optimize.minimize or cvxpy) to find weights that:
 - Maximize Sharpe Ratio
 - Or Minimize Variance for a target return

Part F: Analysis and Visualization

1. Plot the resulting Black-Litterman portfolio weights.
2. Compare with equal-weight portfolio.
3. (Optional) Simulate portfolio performance using cumulative returns.

Submission Guidelines:

- Submit the completed .ipynb notebook.
- Include visualizations and plots.
- Clearly highlight any assumptions made.